Lessons 1 and 2

Discrete random variable

If x is a discrete random variable with pmf $P(X = x)$ with $P(X = x \geq 0), \sum P(X = x) = 1$

Properties of probability mass function pmf

If X is a discrete random variable with P.M.F $P_x(x)$ then

$$P_x(x) \geq 0 \quad \text{i.e } P(X = x) \geq 0$$

$$\sum_{allx} P_x(x) = 1, \sum P(X = x) = 1$$

$$0 \leq P_x(x) \leq 1, \quad 0 \leq P(X = x) \leq 1$$

Example 1

A coin is tossed three times. Let the random variable X be the number of heads that appear. What is the probability distribution of the random variable X?

X={0,1,2,3}

The probability distribution of X becomes

X	0	1	2	3
$P(X = x)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

A die is rolled once. Let the random variable X denote the number facing up. What is the probability mass function p.m.f of X

Solution

$$P(X = x) = P_x(x)$$

But $P(x=1)=\frac{1}{6}, P(x=2)=\frac{1}{6}, \ldots, P(x=6)=\frac{1}{6}$

So $P_x(x)=P(X=x)=\begin{cases}\frac{1}{6}x=1,2,3,4,5,6\\0 \qquad elsewhere\end{cases}$

Continuous random variable

x Is a continuous random variable the function giving the probabilities $f(x)$ is called a probability density function (p.d.f)

Properties for continuous random variable

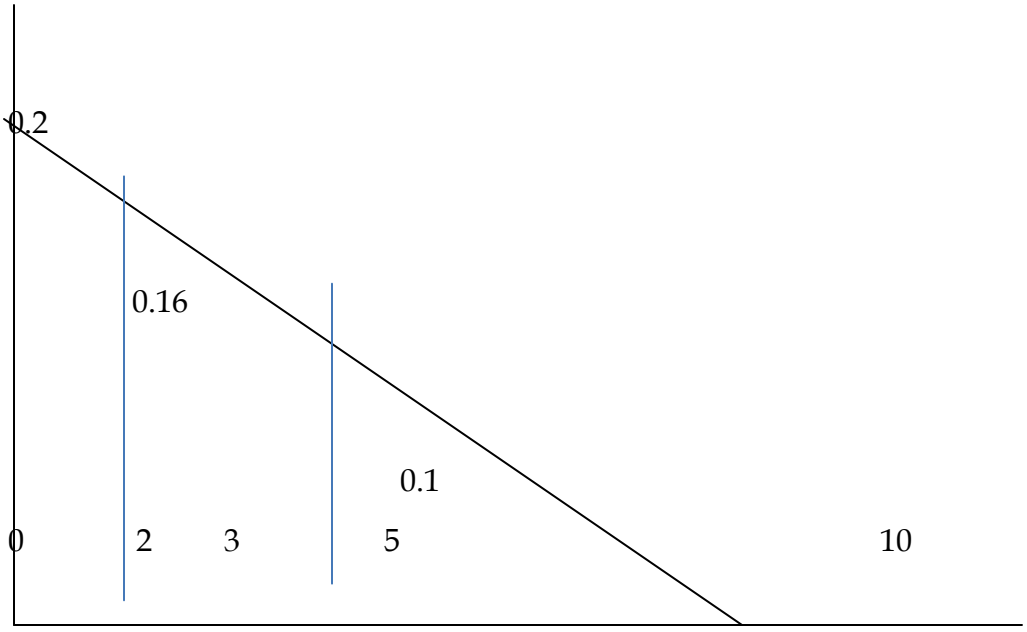
i) $f(x)\geq 0$

ii) $\int f(x)dx=1\ for\ all\ x$

iii) $P(a<x<b)=\int_a^b f(x)dx$

Example

Let x be the delay of hours of a flight where $f(x)=0.2-0.02x \quad 0\leq x\leq 10$



i) $f(x)\geq 0$

ii) Get area under curve = 1 $f(x)$ is a p.d.f

iii) Find $P(x>2)=\frac{1}{2}\times 0.16\times 8=0.64$

iv) Find $P(x \geq 2) = 0.64$

In general for a continuous random variable x with p.d.f $f(x)$ varied over the range $a \leq x \leq b$

Then

1) $\int_a^b f(x)$
2) For $a \leq x_1, b \geq x_2$

$P(x_1 \leq x \leq x_2)$

$= \int_{x_1}^{x_2} f(x)dx$

Example

A continuous random variable has the following p.d.f

$$fx = \begin{cases} k(x+2)^2 & 0 \leq x \leq 2 \\ 0 & \end{cases}$$

i) Find value of K

ii) Find $P(0 < x < 1)$ hence the $P(x > 1)$

Solution

$f(x)$ Is a p.d.f hence

$\int_0^2 f(x) = 1$

$\int_0^2 k(x+2)^2 = 1$

$k(x^2 + 4x + 4)dx$

$\int_0^2 kx^2 + 4kx + 4k$

$\frac{kx^3}{3} + 2kx^2 + 4kx$

$\frac{8k}{3} + 8k + 8k = 1$

$\frac{56}{3}k = 1$

$k = \frac{3}{56}$

$P(0 < x < 1) = \int_0^1 fxdx$

$\frac{3}{56}[\frac{x^3}{3} + 2x^2 + 4x]_0^1 = \frac{19}{56}$

$P(x > 1) = 1 - P(0 < x < 1)$

$= 1 - \frac{19}{56} = \frac{37}{56}$

Example

A p.d.f has $f(x)\begin{cases} k & 0 \le x \le 2 \\ k(2x-3) & 2 \le x \le 3 \\ 0 & otherwise \end{cases}$

i) Find K

ii) Sketch $f(x)$

iii) Find $P(1 < x < 2.5)$

Solution

Since $f(x)$ is a p.d.f then

$\int_0^2 f(x)dx + \int_2^3 f(x)dx = 1$

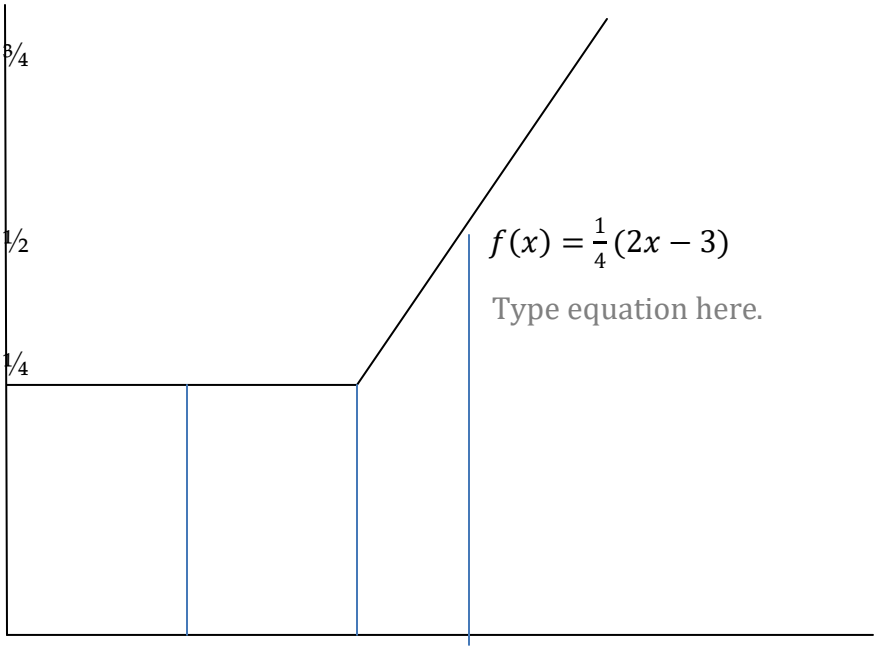
$\int_0^2 k\,dx + k\int_2^3 2x - 3\;dx = 1$

$[x]_0^2 + k[x^2 - 3x]_2^3 = 1$

$k[2 + (0 - -2)] = 1$

$4k = 1$

$k = \frac{1}{4}$



$$P(1 < x < 2.5) = \int_1^2 \frac{1}{4} dx + \frac{1}{4} \int_2^{2.5} 2x - 3 \, dx$$

$$= \frac{1}{4}([x]_1^2 + [x^2 - 3x]_2^{2.5})$$

$$= \frac{1}{4}\left(1 + (-1.25 + 2)\right) = \frac{7}{16}$$